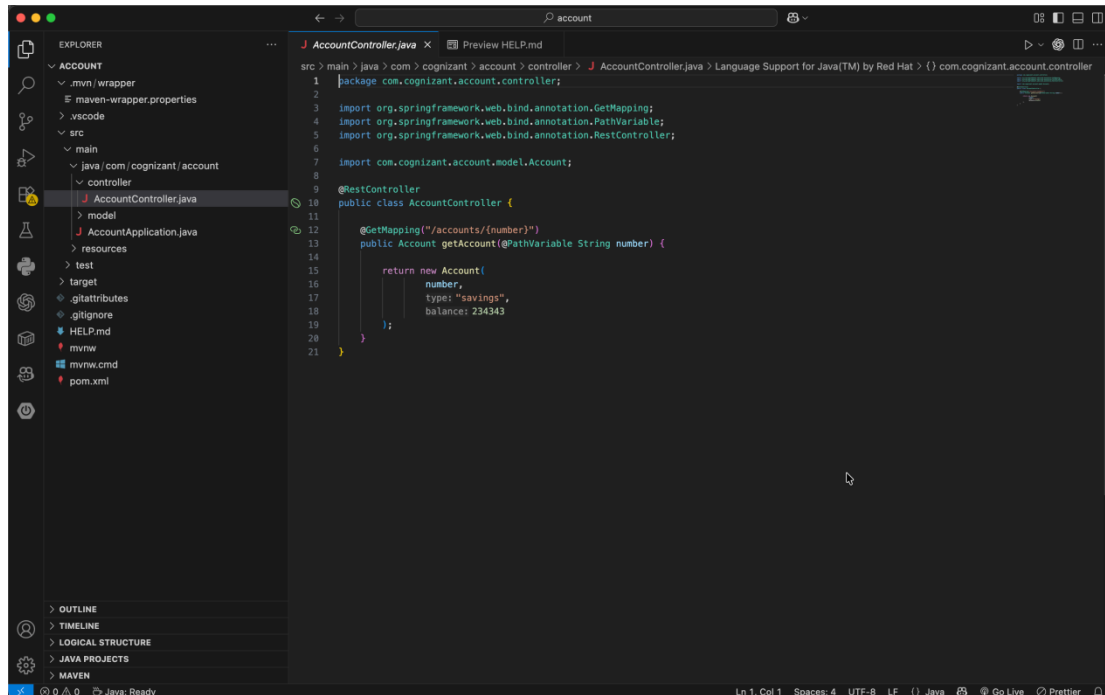


# DIGITAL NURTURE 5.0 DEEPSKILLING

## WEEK 4 – HANDS-ON

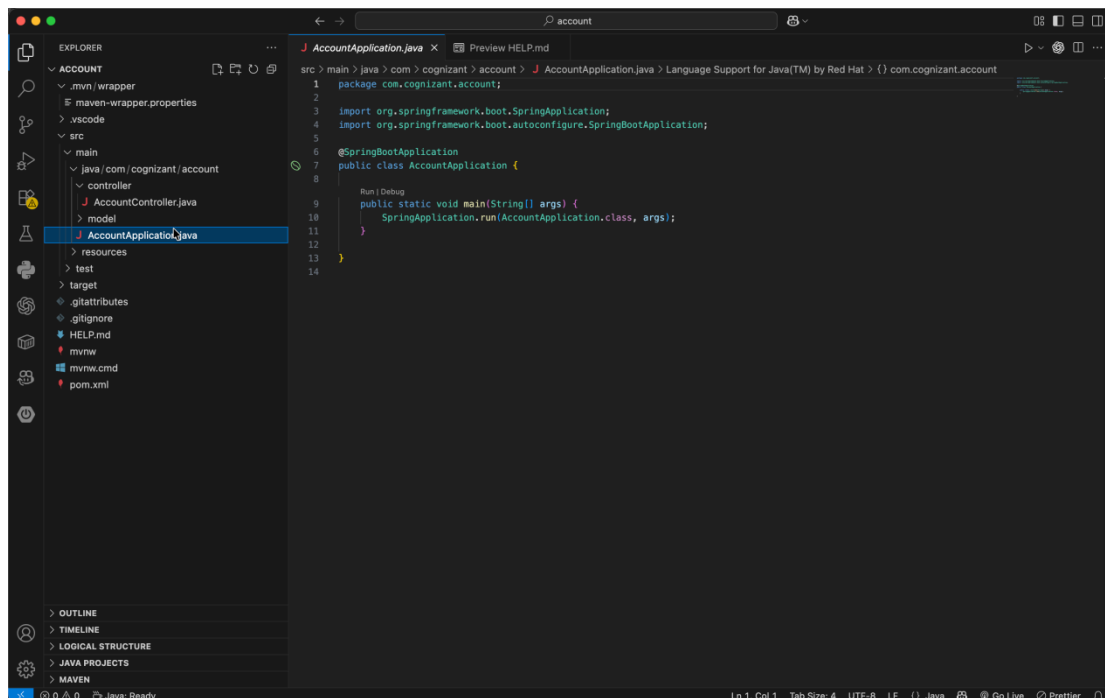
### Micro services with Spring Boot 3 and Spring Cloud

#### ACCOUNT



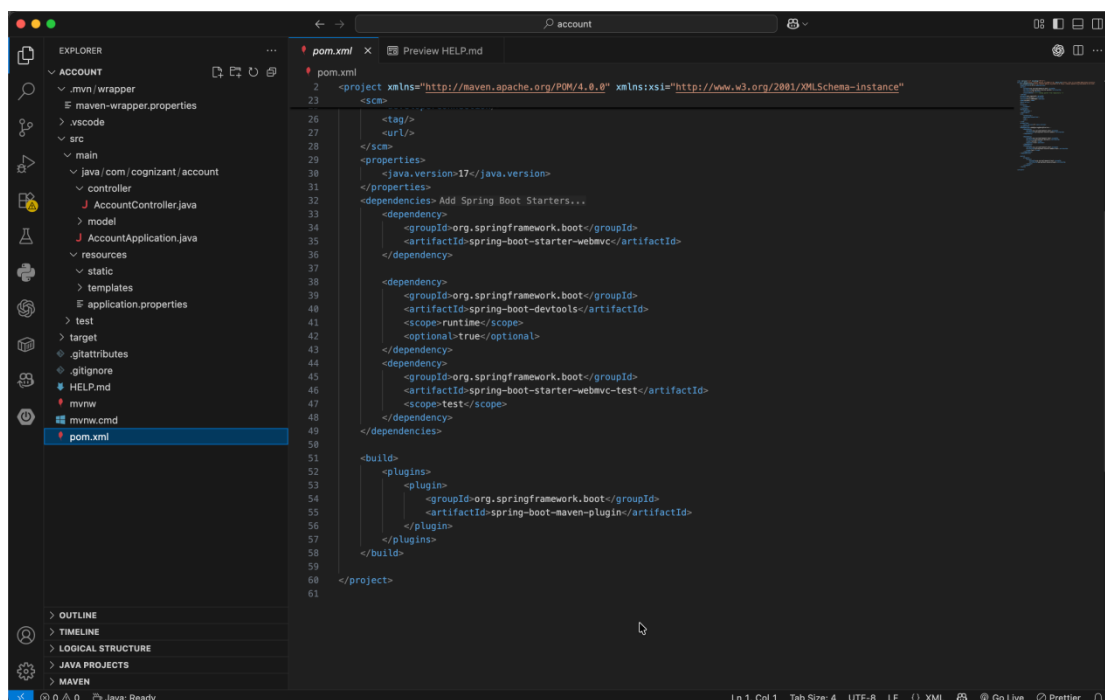
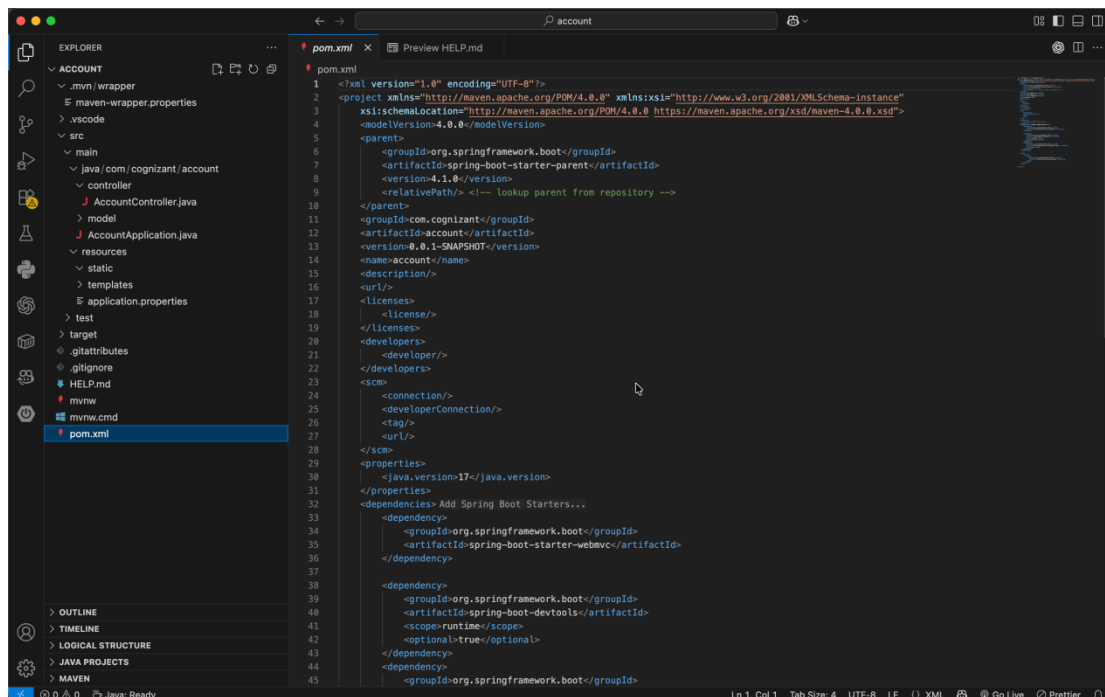
This screenshot shows the Visual Studio Code editor with the `AccountController.java` file open. The Explorer sidebar on the left shows the project structure, including the `controller` directory. The main editor area displays the following Java code:

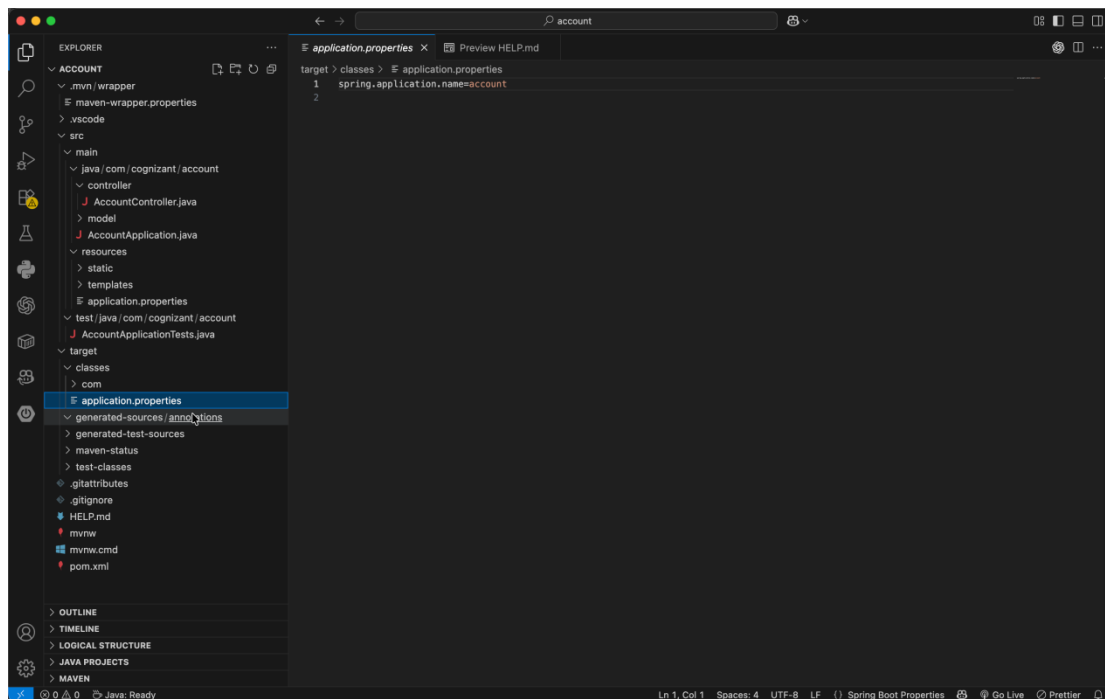
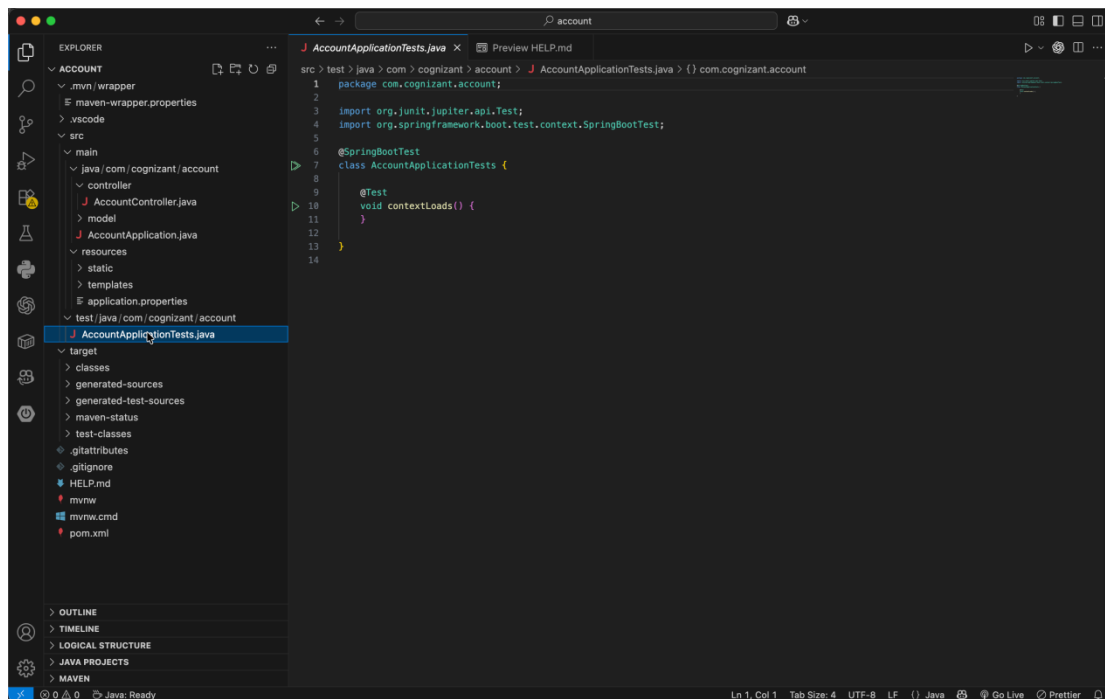
```
src > main > java > com > cognizant > account > controller > J AccountController.java > Language Support for Java(TM) by Red Hat > {} com.cognizant.account.controller  
1 package com.cognizant.account.controller;  
2  
3 import org.springframework.web.bind.annotation.GetMapping;  
4 import org.springframework.web.bind.annotation.PathVariable;  
5 import org.springframework.web.bind.annotation.RestController;  
6  
7 import com.cognizant.account.model.Account;  
8  
9 @RestController  
10 public class AccountController {  
11  
12     @GetMapping("/accounts/{number}")  
13     public Account getAccount(@PathVariable String number) {  
14  
15         return new Account(  
16             number,  
17             type: "savings",  
18             balance: 234343  
19         );  
20  
21     }  
22 }
```



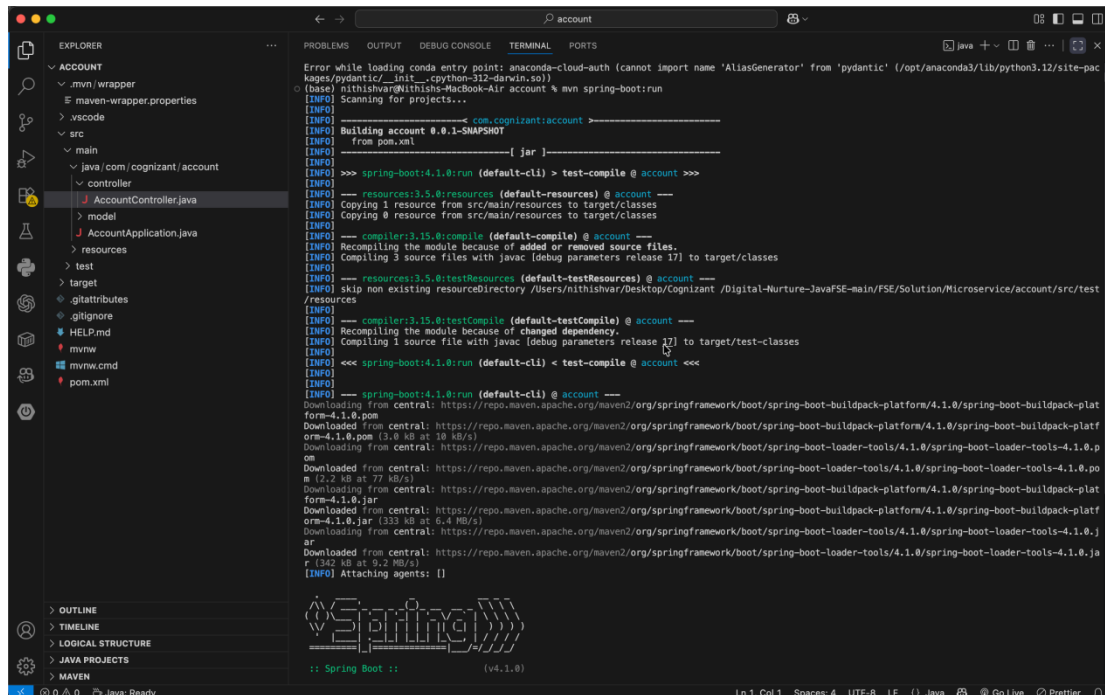
This screenshot shows the Visual Studio Code editor with the `AccountApplication.java` file open. The Explorer sidebar on the left shows the project structure, including the `controller` directory. The main editor area displays the following Java code:

```
src > main > java > com > cognizant > account > J AccountApplication.java > Language Support for Java(TM) by Red Hat > {} com.cognizant.account  
1 package com.cognizant.account;  
2  
3 import org.springframework.boot.SpringApplication;  
4 import org.springframework.boot.autoconfigure.SpringBootApplication;  
5  
6 @SpringBootApplication  
7 public class AccountApplication {  
8  
9     Run | Debug  
10     public static void main(String[] args) {  
11         SpringApplication.run(AccountApplication.class, args);  
12     }  
13  
14 }
```



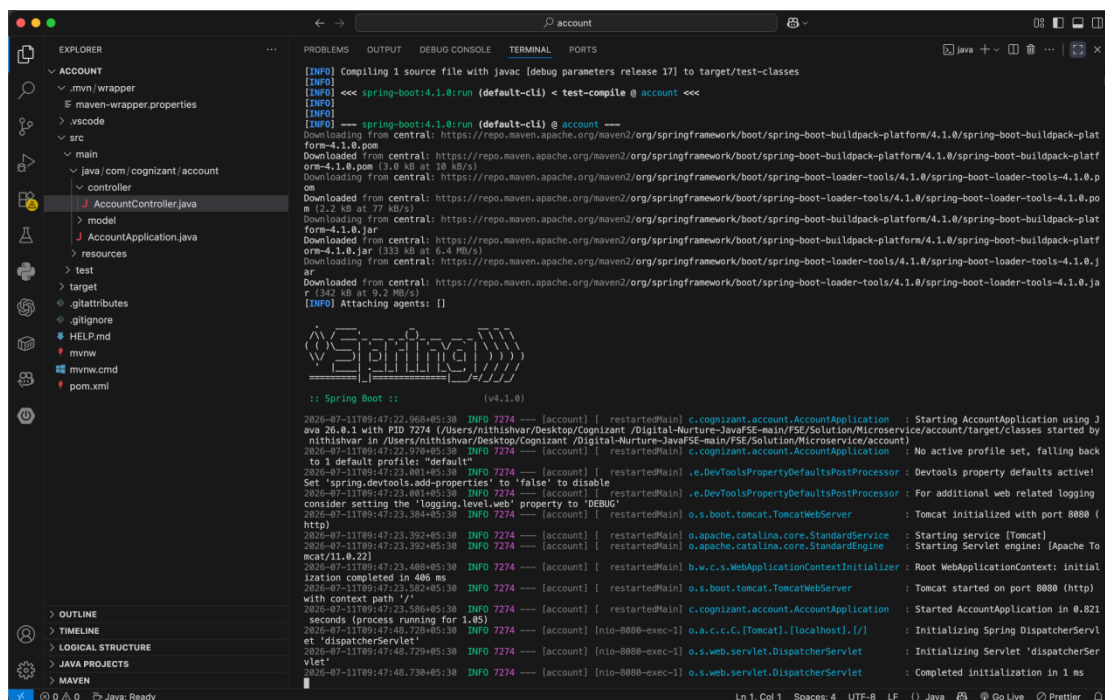


## OUTPUT



```
ERROR while loading conda entry point: anaconda-cloud-auth (cannot import name 'AliasGenerator' from 'pydantic' (/opt/anaconda3/lib/python3.12/site-packages/pydantic/_internal/_cpython_312_darwin.so))
(base) nithishvar@nithishvar-macbook-Air account % mvn spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] <-- com.cognizant:account -->
[INFO]
[INFO] Building account 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] [ jar ]
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ account >>>
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ account ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 4 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ account ---
[INFO] Recompiling the module because of added or removed source files.
[INFO] Compiling 3 source files with javac [debug parameters release 17] to target/classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ account ---
[INFO] skip non existing resourceDirectory /Users/nithishvar/Desktop/Cognizant /Digital-Nurture-JavaFSE-main/FSE/Solution/Microservice/account/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ account ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target/test-classes
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ account <<<
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ account ---
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-buildpack-platform/4.1.0/spring-boot-buildpack-platform-4.1.0.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-buildpack-platform/4.1.0/spring-boot-buildpack-platform-4.1.0.pom (3.0 kB at 10 KB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-loader-tools/4.1.0/spring-boot-loader-tools-4.1.0.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-loader-tools/4.1.0/spring-boot-loader-tools-4.1.0.pom (2.2 kB at 77 KB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-buildpack-platform/4.1.0/spring-boot-buildpack-platform-4.1.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-buildpack-platform/4.1.0/spring-boot-buildpack-platform-4.1.0.jar (333 kB at 6.4 MB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-loader-tools/4.1.0/spring-boot-loader-tools-4.1.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-loader-tools/4.1.0/spring-boot-loader-tools-4.1.0.jar (342 kB at 9.2 MB/s)
[INFO] Attaching agents: []

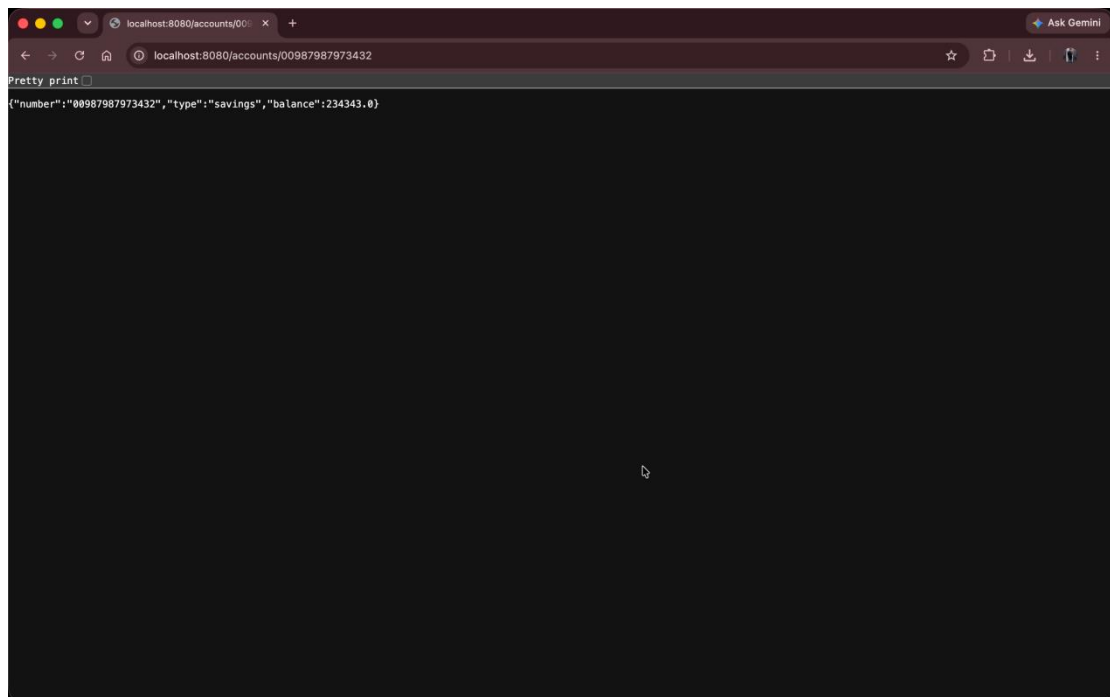
Spring Boot (v4.1.0)
```



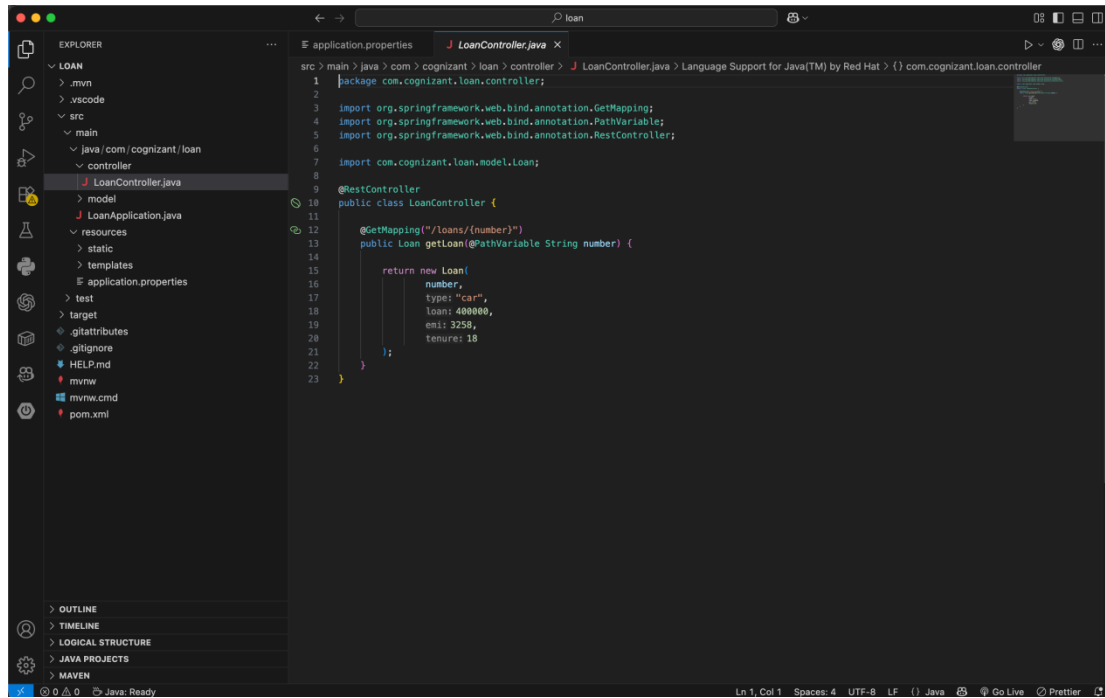
```
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target/test-classes
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ account <<<
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ account ---
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-buildpack-platform/4.1.0/spring-boot-buildpack-platform-4.1.0.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-buildpack-platform/4.1.0/spring-boot-buildpack-platform-4.1.0.pom (3.0 kB at 10 KB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-loader-tools/4.1.0/spring-boot-loader-tools-4.1.0.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-loader-tools/4.1.0/spring-boot-loader-tools-4.1.0.pom (2.2 kB at 77 KB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-buildpack-platform/4.1.0/spring-boot-buildpack-platform-4.1.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-buildpack-platform/4.1.0/spring-boot-buildpack-platform-4.1.0.jar (333 kB at 6.4 MB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-loader-tools/4.1.0/spring-boot-loader-tools-4.1.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-loader-tools/4.1.0/spring-boot-loader-tools-4.1.0.jar (342 kB at 9.2 MB/s)
[INFO] Attaching agents: []

Spring Boot (v4.1.0)

2026-07-11T09:47:22.968+05:30 INFO 7274 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Starting AccountApplication using J
ava 26.0.1 with PID 7274 (/Users/nithishvar/Desktop/Cognizant /Digital-Nurture-JavaFSE-main/FSE/Solution/Microservice/account/target/classes started by
nithishvar in /Users/nithishvar/Desktop/Cognizant /Digital-Nurture-JavaFSE-main/FSE/Solution/Microservice/account)
2026-07-11T09:47:22.978+05:30 INFO 7274 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : No active profile set, falling back
to 1 default profile: "default"
2026-07-11T09:47:23.061+05:30 INFO 7274 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active!
Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-11T09:47:23.061+05:30 INFO 7274 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging
consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-11T09:47:23.304+05:30 INFO 7274 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (
http)
2026-07-11T09:47:23.392+05:30 INFO 7274 --- [account] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-11T09:47:23.392+05:30 INFO 7274 --- [account] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache To
mcat/11.0.22]
2026-07-11T09:47:23.488+05:30 INFO 7274 --- [account] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initial
ization completed in 406 ms
2026-07-11T09:47:23.582+05:30 INFO 7274 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http)
with context path '/'
2026-07-11T09:47:23.585+05:30 INFO 7274 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Started AccountApplication in 0.821
seconds (process running for 1.05)
2026-07-11T09:47:48.728+05:30 INFO 7274 --- [account] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServ
let 'dispatcherServlet'
2026-07-11T09:47:48.729+05:30 INFO 7274 --- [account] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherSer
vlet'
2026-07-11T09:47:48.730+05:30 INFO 7274 --- [account] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
```



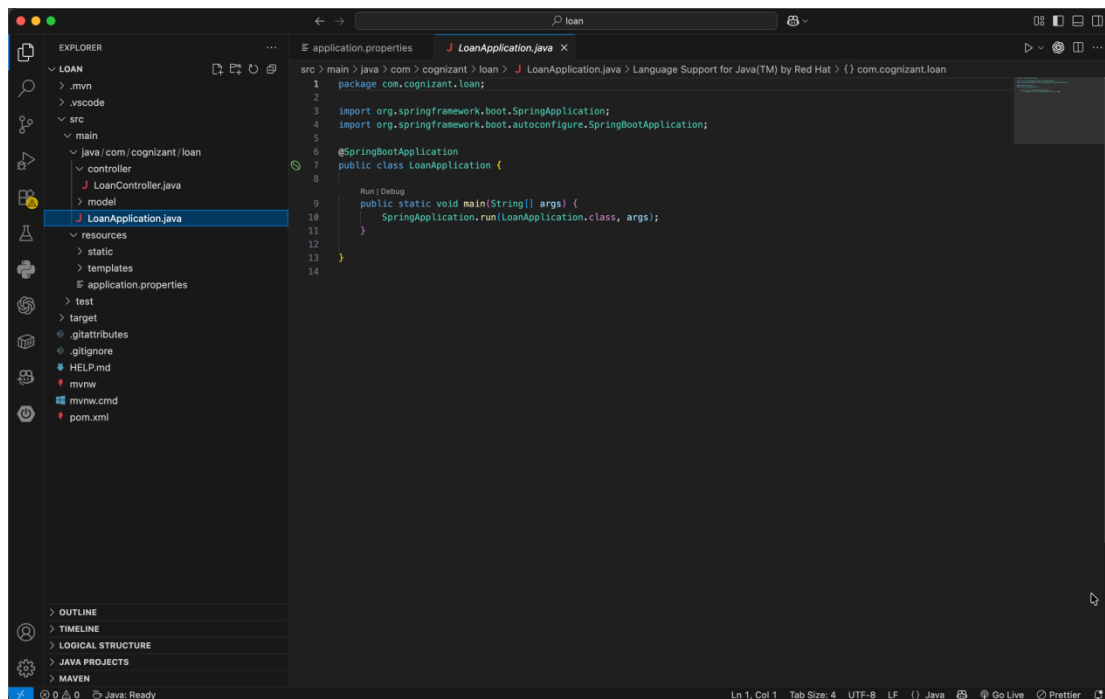
## LOAN



The screenshot shows the VS Code editor with the `LoanController.java` file open. The Explorer sidebar on the left shows the project structure: `LOAN` (root) contains `.mvn`, `.vscode`, `src` (main, test), `target`, `.gitattributes`, `.gitignore`, `HELP.md`, `mvnw`, `mvnw.cmd`, and `pom.xml`. The `src/main/java/com/cognizant/loan` package contains `controller` (with `LoanController.java`), `model` (with `LoanApplication.java`), `resources` (static, templates), `application.properties`, and `test`. The `LoanController.java` file is a REST controller with the following code:

```
src > main > java > com > cognizant > loan > controller > J LoanController.java > Language Support for Java(TM) by Red Hat > {} com.cognizant.loan.controller
1 package com.cognizant.loan.controller;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.PathVariable;
5 import org.springframework.web.bind.annotation.RestController;
6
7 import com.cognizant.loan.model.Loan;
8
9 @RestController
10 public class LoanController {
11
12     @GetMapping("/loans/{number}")
13     public Loan getLoan(@PathVariable String number) {
14
15         return new Loan(
16             number,
17             type: "car",
18             loan: 400000,
19             emi: 3258,
20             tenure: 18
21         );
22     }
23 }
```

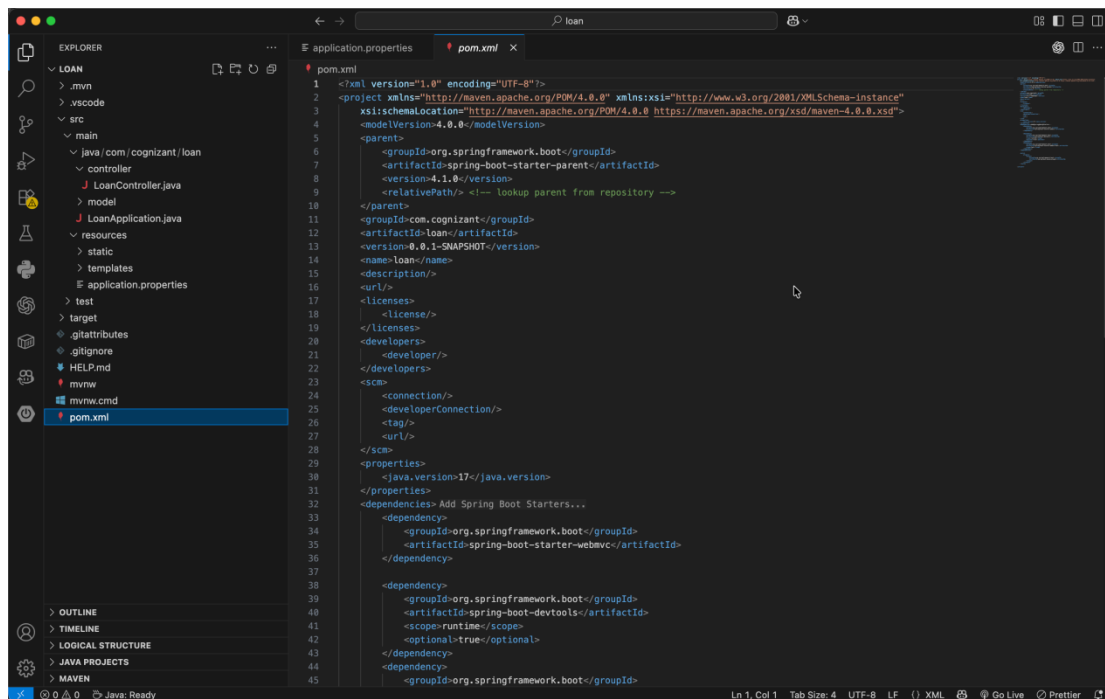
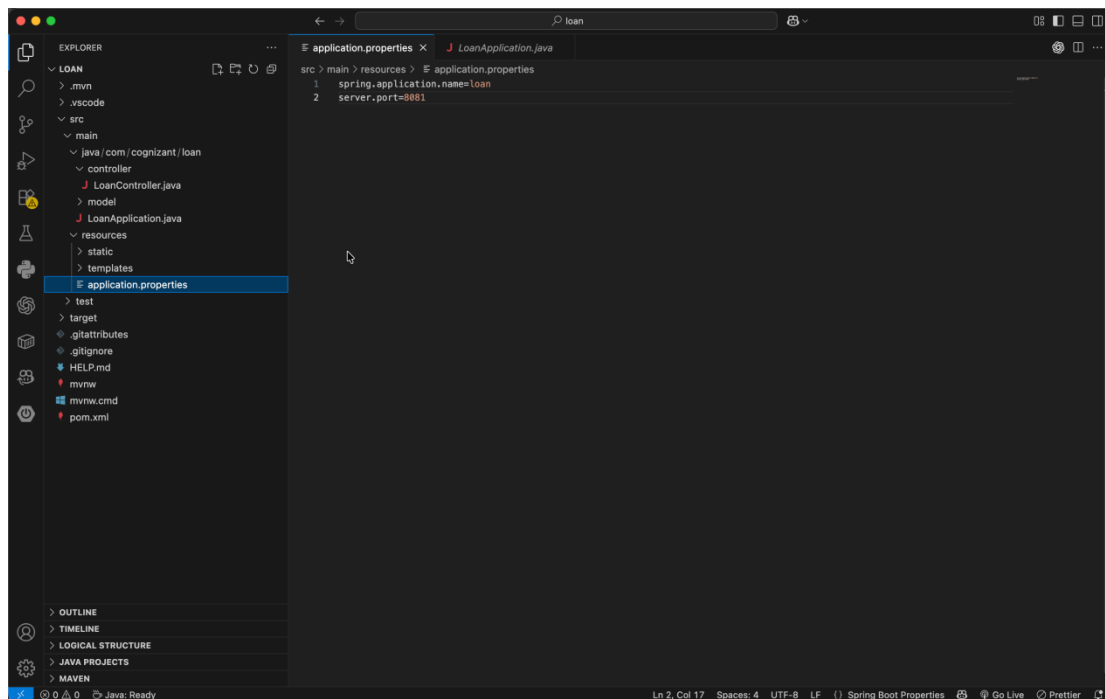
The status bar at the bottom indicates the file is at line 1, column 1, with 4 spaces, UTF-8 encoding, LF line endings, and is a Java file. It also shows the Go Live and Prettier icons.

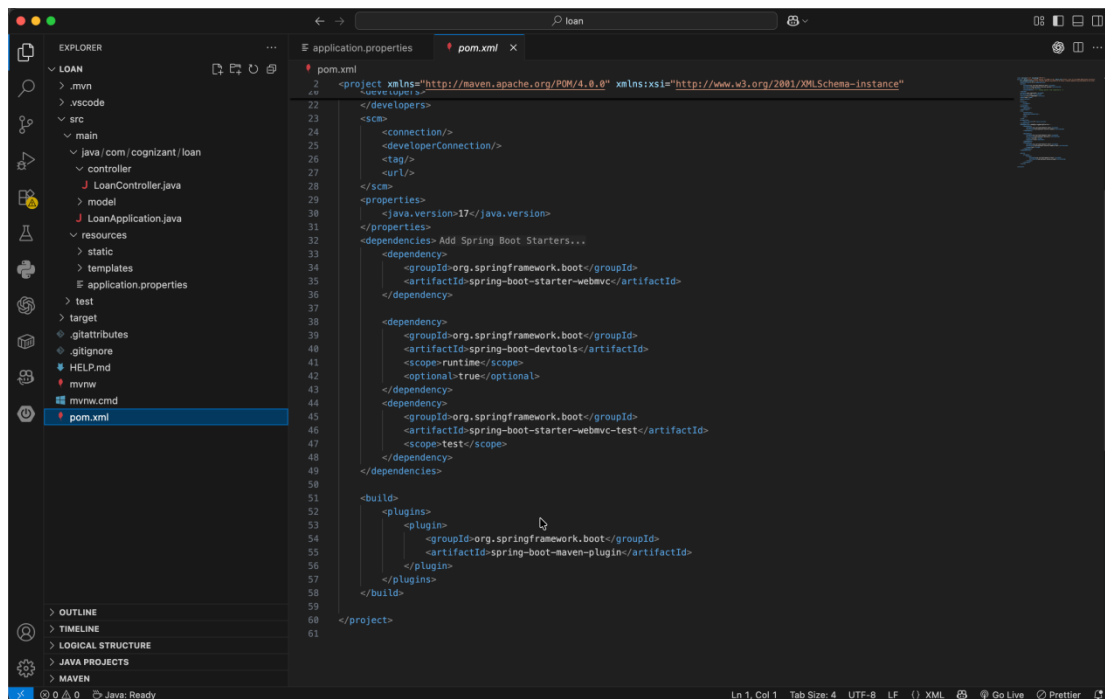


The screenshot shows the VS Code editor with the `LoanApplication.java` file open. The Explorer sidebar on the left shows the project structure: `LOAN` (root) contains `.mvn`, `.vscode`, `src` (main, test), `target`, `.gitattributes`, `.gitignore`, `HELP.md`, `mvnw`, `mvnw.cmd`, and `pom.xml`. The `src/main/java/com/cognizant/loan` package contains `controller` (with `LoanController.java`), `model` (with `LoanApplication.java`), `resources` (static, templates), `application.properties`, and `test`. The `LoanApplication.java` file is a Spring Boot application with the following code:

```
src > main > java > com > cognizant > loan > J LoanApplication.java > Language Support for Java(TM) by Red Hat > {} com.cognizant.loan
1 package com.cognizant.loan;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class LoanApplication {
8
9     Run | Debug
10     public static void main(String[] args) {
11         SpringApplication.run(LoanApplication.class, args);
12     }
13 }
14 }
```

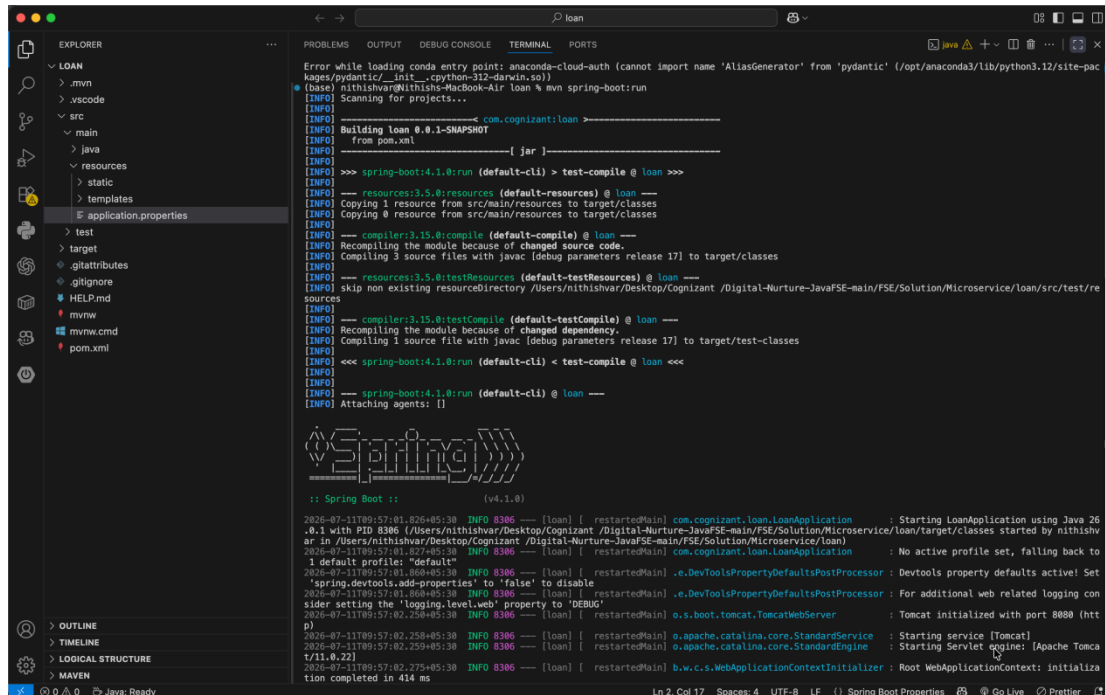
The status bar at the bottom indicates the file is at line 1, column 1, with a tab size of 4, UTF-8 encoding, LF line endings, and is a Java file. It also shows the Go Live and Prettier icons.







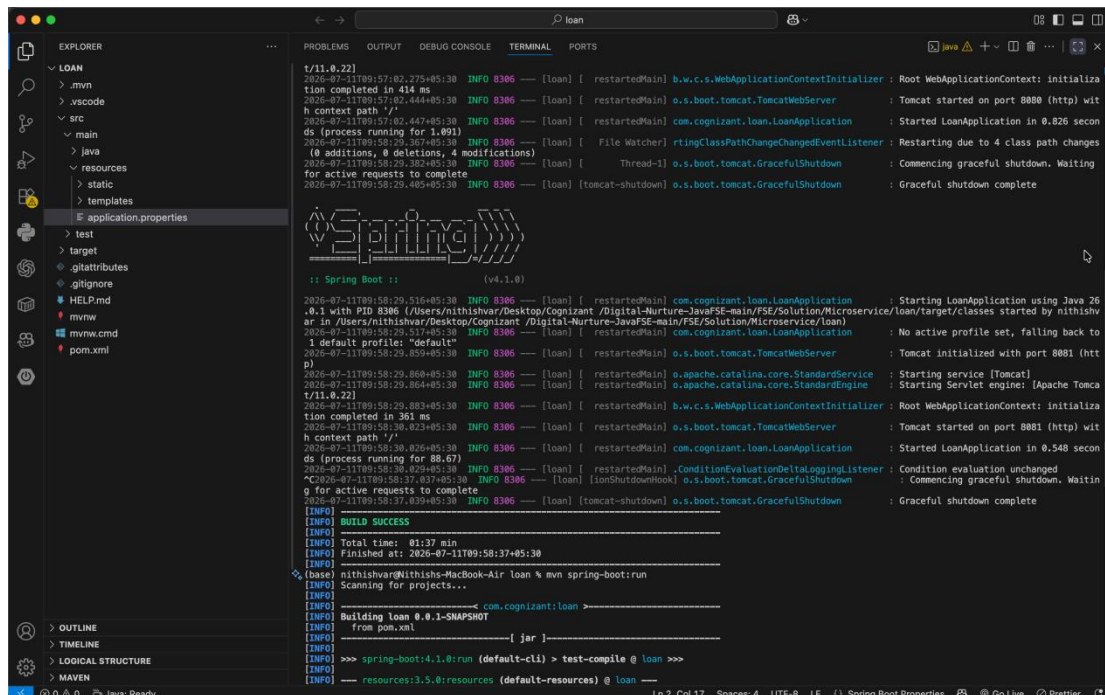
## OUTPUT



```
Error while loading conda entry point: anaconda-cloud-auth (cannot import name 'AliasGenerator' from 'pydantic' (/opt/anaconda3/lib/python3.12/site-packages/pydantic/_internal/_cpython_312_darwin.so))
(base) nithishvar@Nithishs-MacBook-Air loan % mvn spring-boot:run
[INFO] Scanning for projects...
[INFO] Building loan 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] [ jar ]
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ loan >>>
[INFO] --- resources:3.5.0:resources (default-resources) @ loan ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO] --- compiler:3.15.0:compile (default-compile) @ loan ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 3 source files with javac [debug parameters release 17] to target/classes
[INFO] --- resources:3.5.0:testResources (default-testResources) @ loan ---
[INFO] skip non existing resourceDirectory /Users/nithishvar/Desktop/Cognizant /Digital-Nurture-JavaFSE-main/FSE/Solution/Microservice/loan/src/test/resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ loan ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target/test-classes
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ loan <<<
[INFO] --- spring-boot:4.1.0:run (default-cli) @ loan ---
[INFO] Attaching agents: []

Spring Boot (v4.1.0)

2026-07-11T09:57:01.826+05:30 INFO 8306 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Starting LoanApplication using Java 26
0.1 with PID 8306 (/Users/nithishvar/Desktop/Cognizant /Digital-Nurture-JavaFSE-main/FSE/Solution/Microservice/loan/target/classes started by nithishv
ar in /Users/nithishvar/Desktop/Cognizant /Digital-Nurture-JavaFSE-main/FSE/Solution/Microservice/loan)
2026-07-11T09:57:01.827+05:30 INFO 8306 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : No active profile set, falling back to
1 default profiles: "default"
2026-07-11T09:57:01.868+05:30 INFO 8306 --- [loan] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set
'spring.devtools.add-properties' to 'false' to disable
2026-07-11T09:57:01.868+05:30 INFO 8306 --- [loan] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging con
sider setting the 'logging.level.web' property to 'DEBUG'
2026-07-11T09:57:02.250+05:30 INFO 8306 --- [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (htt
p)
2026-07-11T09:57:02.250+05:30 INFO 8306 --- [loan] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-11T09:57:02.250+05:30 INFO 8306 --- [loan] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomca
t]
2026-07-11T09:57:02.275+05:30 INFO 8306 --- [loan] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initializa
tion completed in 414 ms
Ln 2, Col 17 Spaces: 4 UTF-8 LF ( ) Spring Boot Properties @ Go Live Prettier
```



```
t/11.0.22]
2026-07-11T09:57:02.275+05:30 INFO 8306 --- [loan] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initializa
tion completed in 414 ms
2026-07-11T09:57:02.444+05:30 INFO 8306 --- [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) wit
h context path '/'
2026-07-11T09:57:02.447+05:30 INFO 8306 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Started LoanApplication in 0.826 secon
ds (process running for 1.001)
2026-07-11T09:58:29.367+05:30 INFO 8306 --- [loan] [ File Watcher] rtingClassPathChangeChangedEventListener : Restarting due to 4 class path changes
(0 additions, 0 deletions, 4 modifications)
2026-07-11T09:58:29.367+05:30 INFO 8306 --- [loan] [ Thread-1] o.s.boot.tomcat.GracefulShutdown : Commencing graceful shutdown. Waiting
for active requests to complete
2026-07-11T09:58:29.405+05:30 INFO 8306 --- [loan] [tomcat-shutdown] o.s.boot.tomcat.GracefulShutdown : Graceful shutdown complete

Spring Boot (v4.1.0)

2026-07-11T09:58:29.516+05:30 INFO 8306 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Starting LoanApplication using Java 26
0.1 with PID 8306 (/Users/nithishvar/Desktop/Cognizant /Digital-Nurture-JavaFSE-main/FSE/Solution/Microservice/loan/target/classes started by nithishv
ar in /Users/nithishvar/Desktop/Cognizant /Digital-Nurture-JavaFSE-main/FSE/Solution/Microservice/loan)
2026-07-11T09:58:29.517+05:30 INFO 8306 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : No active profile set, falling back to
1 default profiles: "default"
2026-07-11T09:58:29.559+05:30 INFO 8306 --- [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (htt
p)
2026-07-11T09:58:29.860+05:30 INFO 8306 --- [loan] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-11T09:58:29.864+05:30 INFO 8306 --- [loan] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomca
t]
t/11.0.22]
2026-07-11T09:58:29.863+05:30 INFO 8306 --- [loan] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initializa
tion completed in 361 ms
2026-07-11T09:58:30.023+05:30 INFO 8306 --- [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) wit
h context path '/'
2026-07-11T09:58:30.026+05:30 INFO 8306 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Started LoanApplication in 0.548 secon
ds (process running for 86.67)
2026-07-11T09:58:30.829+05:30 INFO 8306 --- [loan] [ restartedMain] .ConditionEvaluationDeltaLoggingListener : Condition evaluation unchanged
2026-07-11T09:58:37.037+05:30 INFO 8306 --- [loan] [onShutdownHook] o.s.boot.tomcat.GracefulShutdown : Commencing graceful shutdown. Waitin
g for active requests to complete
2026-07-11T09:58:37.039+05:30 INFO 8306 --- [loan] [tomcat-shutdown] o.s.boot.tomcat.GracefulShutdown : Graceful shutdown complete

BUILD SUCCESS
[INFO] Total time: 01:37 min
[INFO] Finished at: 2026-07-11T09:58:37+05:30
[INFO] Scanning for projects...
[INFO] Building loan 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] [ jar ]
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ loan >>>
[INFO] --- resources:3.5.0:resources (default-resources) @ loan ---
(base) nithishvar@Nithishs-MacBook-Air loan % mvn spring-boot:run
[INFO] Scanning for projects...
[INFO] Building loan 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] [ jar ]
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ loan >>>
[INFO] --- resources:3.5.0:resources (default-resources) @ loan ---
Ln 2, Col 17 Spaces: 4 UTF-8 LF ( ) Spring Boot Properties @ Go Live Prettier
```



